

Data Visualization Mastery with Seaborn sProject Report

Project Overview

The Interactive Sales Dashboard analyzes sales data using Python and presents business insights through Seaborn, Matplotlib, and Plotly visualizations.

Objectives

- Analyze sales data
- Create statistical and interactive charts
- Generate business insights
- Save visualizations automatically

Dataset

File: sales_data.csv

Fields: Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales

Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter Notebook, VS Code

Setup and Installation

1. Install Python
2. Run: `pip install -r requirements.txt`
3. Execute `dashboard.py` or `dashboard.ipynb`

Code Structure

`dashboard.py`: main program

`dashboard.ipynb`: notebook version

`visualizations/`: saved charts

Dashboard Guide

Line Chart: Sales trend

Bar Chart: Product sales

Scatter Plot: Price vs Sales

Box Plot: Price distribution

Heatmap: Correlation analysis

Plotly: Interactive exploration

Technical Requirements

Used Seaborn, created 5 chart types, included Plotly interactive charts, generated business insights, saved visualizations.

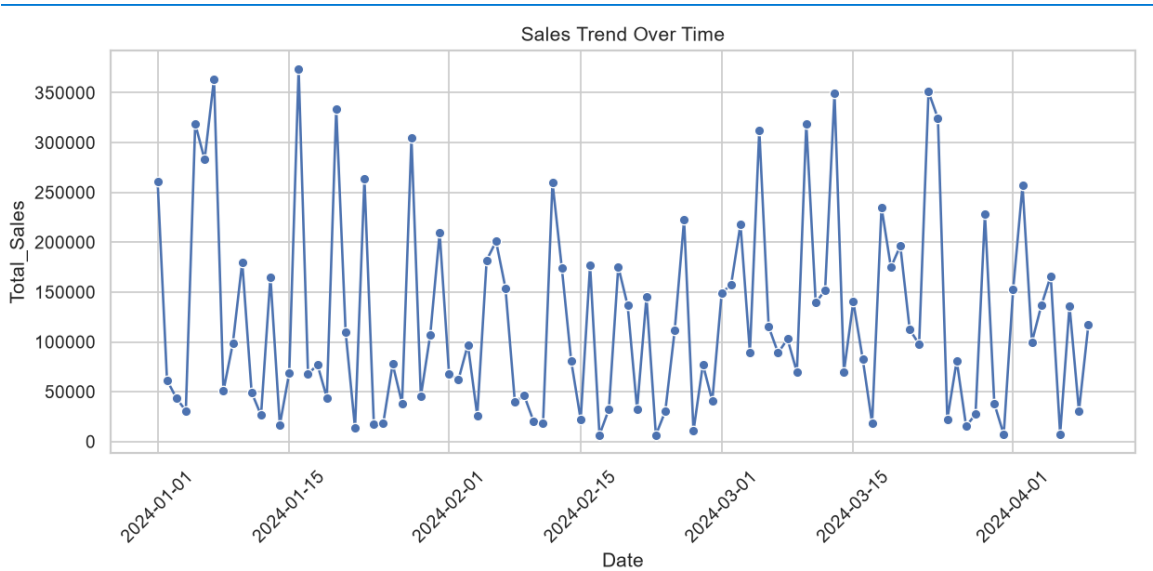
Screenshots

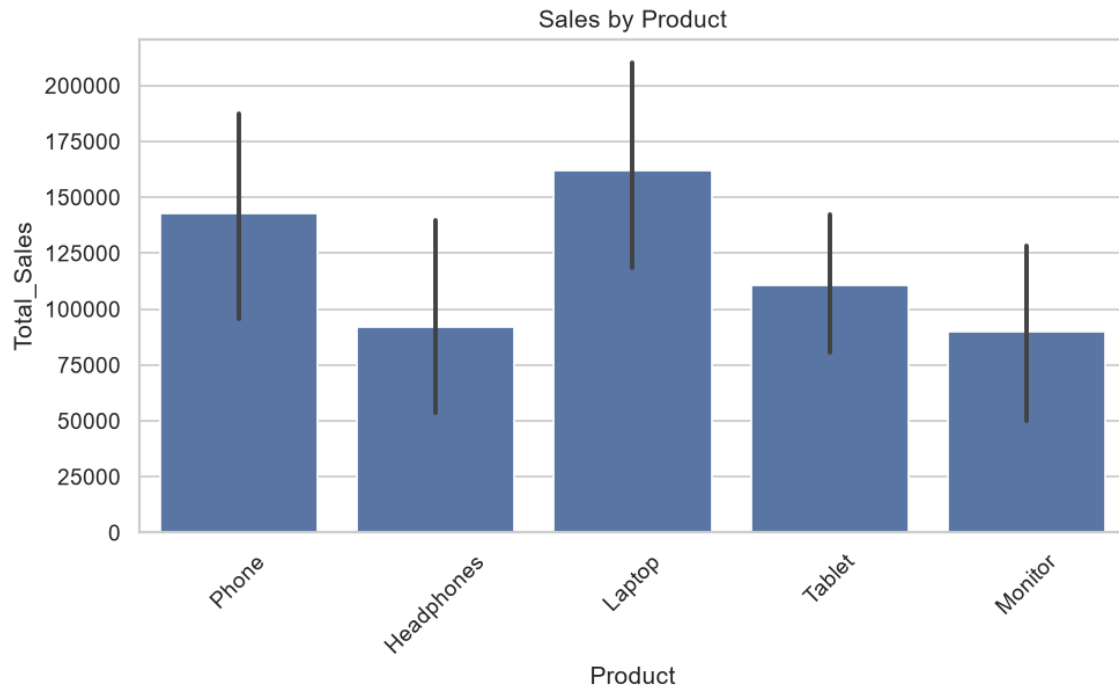
Insert screenshots of dashboard and charts here before submission.

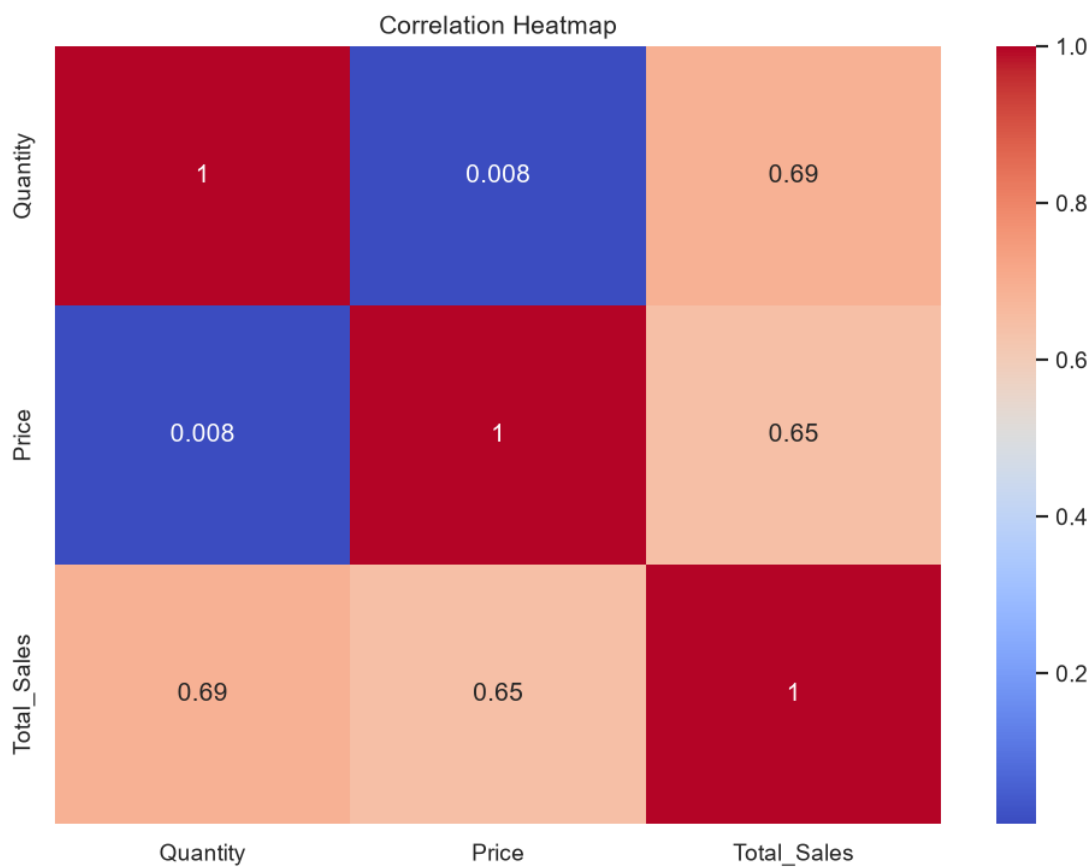
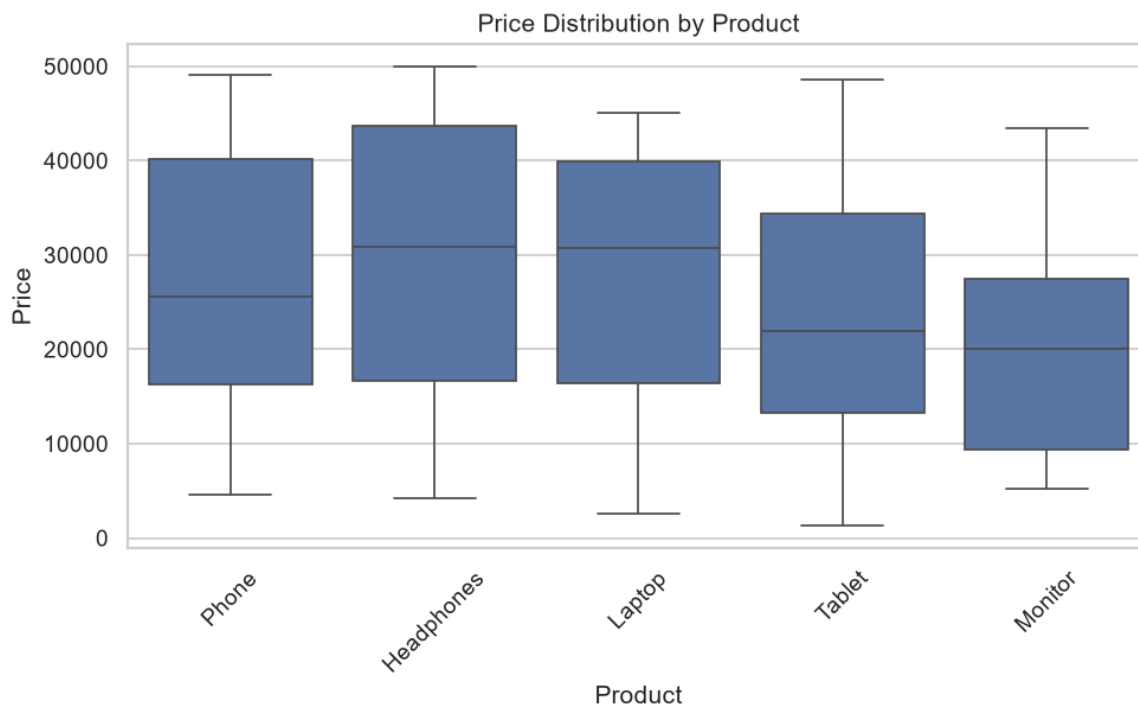
Conclusion

The project demonstrates effective sales analysis using professional data visualization techniques.

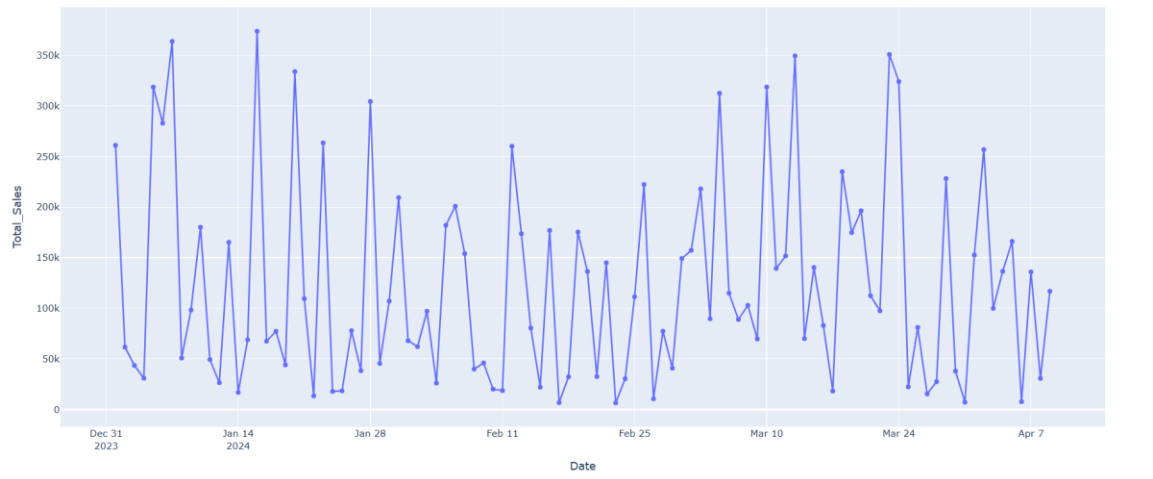
Output:



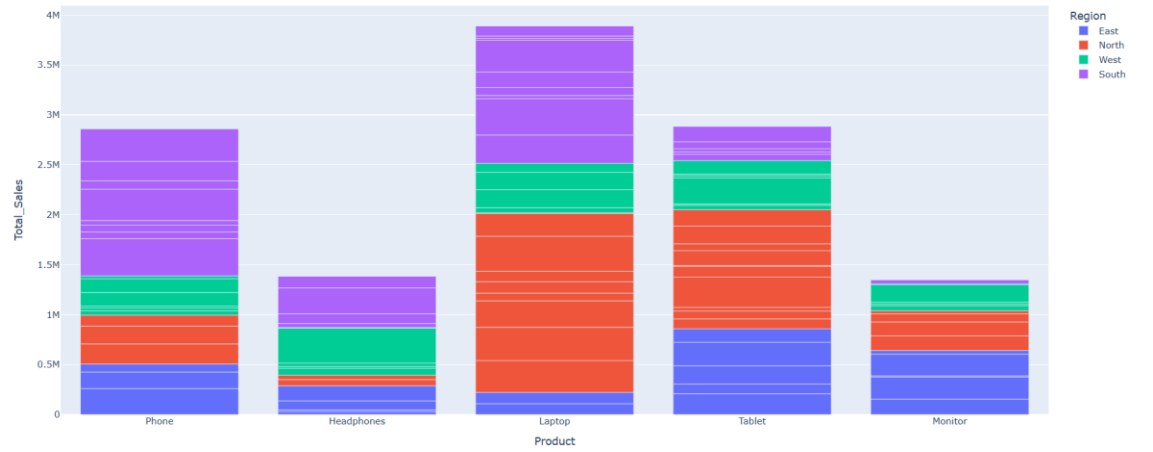




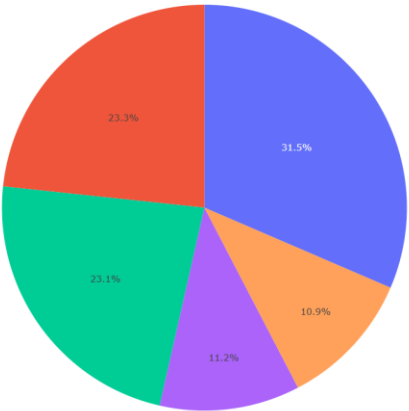
Interactive Sales Trend



Interactive Sales by Product



Sales Share by Product



- Laptop
- Tablet
- Phone
- Headphones
- Monitor